Neelkumar K. Shah

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https://neelkumarkshahportfolio.vercel.app

PROFESSIONAL SUMMARY

Engineer with 5 years of experience in blockchain development, full-stack engineering, and cybersecurity. Built and hardened Hyperledger Fabric and Ethereum systems for India's National Blockchain Service initiative under the Ministry of Electronics and Information Technology (MeitY). Developed a full-stack security assessment platform (React, Node.js, Express.js, MongoDB) and an automated database vulnerability scanner. Authored 5 peer-reviewed IEEE publications on blockchain security.

SKILLS

Blockchain: Hyperledger (Fabric, Sawtooth), Ethereum, Solidity, Truffle, Ganache, IPFS, MetaMask
Development: Python, Node.js, Express.js, NestJS, React, MongoDB, HTML/CSS, Bootstrap
Security: Vulnerability Assessment & Penetration Testing (Web Applications, APIs), Smart Contract Audits (Ethereum, Hyperledger), Static Source Code Analysis, NoSQL Database Security Audit

PROFESSIONAL EXPERIENCE

Self-Employed/Freelance

Independent Full-Stack Developer, Blockchain Engineer & Cybersecurity Consultant | Aug 2024 - Present

- Architected and developed Sudarshana, a full-stack web application security auditing platform built with Node.js, React.js, Express.js, and MongoDB (MERN stack) to automate security assessments for web applications and REST APIs.
- Designed and developed an automated Node.js security scanner that analyzes web applications and APIs for HTTP security headers.
- Developed robust RESTful API endpoints, secure authentication workflows, and interactive React.js dashboards with automated PDF audit report generation, enabling real-time security posture assessment, vulnerability tracking, and remediation management for scanned web applications and API endpoints.
- Developed Solidity smart contracts for Ethereum-based decentralized applications and performed source code security analysis of Solidity smart contracts and Node.js chaincode to identify vulnerabilities, logic flaws, and compliance gaps.

School of Applied Sciences, Engineering and Technology - Rashtriya Raksha University

Assistant Professor | Gandhinagar, Gujarat, India | Jun 2026 - Aug 2026

- Delivered a full-day training on blockchain and cryptocurrency fundamentals to 20 NCB officers and technical staff from across India.
- Authored Board of Studies (BoS) documentation establishing the curriculum for a new PG Diploma in Blockchain Technology program.

C-DAC: Centre for Development of Advanced Computing

Project Engineer | Pune, Maharashtra, India | Oct 2021 - Jun 2026

- Led the MeitY-funded initiative to build the National Blockchain Framework (NBF) for India's National Blockchain Service - full technical build detailed under Projects.
- Co-led a 7-person team building a full-stack Blockchain Security Assessment Portal (React, Node.js, Express, MongoDB) automating evaluations across 10-15 APIs, apps, containers, and databases.

Huptech Web Private Limited

Jr. UI Designer and Developer | Ahmedabad, Gujarat, India | Jun 2019 - Sep 2019

- Built responsive front-end interfaces using HTML, CSS, JavaScript, jQuery, and Bootstrap.

PROJECTS

National Blockchain Framework (NBF) - National Blockchain Service Initiative, MeitY

C-DAC: Centre for Development of Advanced Computing | Oct 2021 - Jun 2026

- Designed and architected a unified, multi-organization Hyperledger Fabric framework (3 peer nodes, 2 organizations, 1 channel) across 2-3 test environments.
- Built REST APIs for a security audit integrating blockchain services.
- Designed and built NBFDBAudit, a Python-based scanner that detects 18+ vulnerability classes across 4-5 MongoDB/CouchDB environments (including 1 production).
- Led VAPT across 4-5 blockchain applications and 30-45 APIs, surfacing ~25 vulnerabilities per application on average; partnered with engineering on remediation.
- Authored and presented 5 peer-reviewed papers at IEEE ICBDS and I2CT conferences on blockchain and smart contract security.

Development and Deployment of E-waste Management Modules Using Blockchain Technology

Graduate School of Engineering and Technology, GTU (M.E. Dissertation) | Jun 2020 - Jul 2021

- Designed and built a decentralized Ethereum DApp using Solidity, Truffle, Ganache, and IPFS. Implemented secure, tamper-resistant ownership tracking across the e-waste lifecycle.
- Integrated a React.js frontend with MetaMask wallet connectivity for on-chain interactions.
- Conducted threat modelling and optimized the decentralized architecture for performance.

PUBLICATIONS

A Threat Modeling Approach for Securing Blockchain Framework - 2025 4th IEEE ICBDS

Security Assessment Framework and Evaluation for Blockchain Applications (SAFE-Block) - 2024 3rd IEEE ICBDS

Efficient Solution for NoSQL Database Security in Blockchain-Based Applications - 2023 2nd IEEE ICBDS

Smart Contract Vulnerability Detection Techniques for Hyperledger Fabric - 2023 8th IEEE I2CT

Smart Contract - Security Assessment Integrated Framework (SC-SIF) for Hyperledger Fabric - 2022 7th IEEE I2CT

EDUCATION

Master's Degree in Cybersecurity

Graduate School of Engineering and Technology, GTU, Ahmedabad, Gujarat, India | Sep 2019 - Jul 2021

Bachelor's Degree in Computer Engineering

Marwadi Education Foundation, Rajkot, Gujarat, India | Aug 2015 - May 2019

CERTIFICATION & ACHIEVEMENTS

- Certified Blockchain Security Professional - Blockchain Council, USA
- Recognized as Performer of the Month - C-DAC, Pune, Maharashtra, India
- Blockchain Technology for Government Officials - Future Skills PRIME, NIELIT, Kolkata, India